

4.1 Hindu-Arabic Positional System

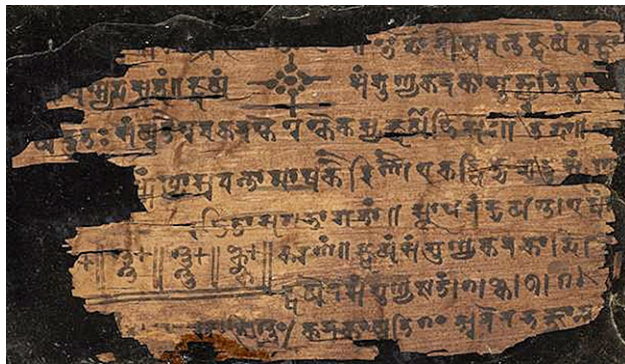


Figure 4.2 This manuscript is an early example of Hindu numerals. (credit: modification of work “Bakshali manuscript”, Bodleian Libraries/ University of Oxford, public domain)

Learning Objectives

After completing this section, you should be able to:

1. Evaluate an exponential expression.
2. Convert a Hindu-Arabic numeral to expanded form.
3. Convert a number in expanded form to a Hindu-Arabic numeral.

The modern system of counting and computing isn't necessarily natural. That different symbols are used to indicate different quantities or amounts is a relatively new invention. Simple marking by scratches or dots, one for each item being counted, was the norm long into human history. The modern system doesn't use repeated symbols to indicate more than one of a thing. It uses the place of a digit in a **numeral** to determine what that digit represents. A numeral is a symbol used to represent a number. A **number** is an abstract idea that represents quantity or amount.

Being clear about the difference between numeral and number is important. Just like a person can be called by various names, such as brother, father, husband, uncle, they are all representing the same person, John Smith. The person John Smith is the number, and the names brother, father, husband, and uncle are the numerals.

? WHO KNEW?

Hindu-Arabic Numerals

The numerals we currently use are referred to as Hindu-Arabic numerals, although they have changed as time has passed. Early forms of the numerals for 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9 began in India, and passed through Persia to the Middle East. Place value was also employed in the early systems of India. Once this system was in north Africa and the Middle East, it spread to Europe, eventually replacing Roman numerals. Over time, the original symbols transformed into our modern ones. Read [this article for another perspective on how the symbols began \(based on the moon!\)](https://openstax.org/r/myindiayglory.com/2018/09/05) (<https://openstax.org/r/myindiayglory.com/2018/09/05>).

The system we use for counting and computing uses place values based on powers of 10. In this section, we review exponents and our positional system.

Evaluating Exponential Expressions

Most modern numerical systems depend on place values, where the quantity represented depends not only on the digit, but also on where the digit is in the number. The place value is a power of some specific number, which means most numbering systems are actually exponential expressions. An **exponential expression** is any mathematical expression that includes exponents. So, evaluating such an expression means performing the calculation. In this chapter, we will be using exponents that are positive integer values. Before we do so, let's remind ourselves about exponents and what they represent. Suppose you want to multiply a number. Let's label that number a , by itself some number of times. Let's label the number of times b . We denote that as a^b . We say a , or the **base**, raised to the b th power, or the **exponent**. For example, if we are multiplying 13 by itself eight times, we write 13^8 and say 13 to the eighth power.

When computing exponential expressions, we should be careful to remember the order of operations. Using the order of operation rules, calculations inside the parentheses are done first, then exponents are calculated, then multiplication

and division calculations are performed, and then addition and subtraction.

 VIDEO

[Exponential Notation \(https://openstax.org/r/Exponential_Notation\)](https://openstax.org/r/Exponential_Notation)

EXAMPLE 4.1

Evaluating an Exponential Expression

Evaluate the following exponential expressions.

1. $4 \times 5^2 + 2 \times 6^3$
2. $6 \times 8^2 + 3 \times 8^1 + 4 \times 8^0$
3. $3 \times 10^2 + 0 \times 10^1 + 6 \times 10^0$

 Solution

1. To evaluate, or calculate, this expression, we use order of operations, which means the exponents are done first, then multiplications, and then additions.

$$4 \times 5^2 + 2 \times 6^3 = 4 \times 5 \times 5 + 2 \times 6 \times 6 \times 6 = 4 \times 25 + 2 \times 216 = 100 + 432 = 532$$

2. To evaluate the expression, we use the order of operations, which means the exponents are done first, then the multiplications, then the additions. Remember that any base raised to the exponent 0 is 1.

$$6 \times 8^2 + 3 \times 8^1 + 4 \times 8^0 = 6 \times 8 \times 8 + 3 \times 8 + 4 \times 1 = 6 \times 64 + 3 \times 8 + 4 \times 1 = 384 + 24 + 4 = 412$$

3. To evaluate the expression, we use the order of operations, which means the exponents are done first, then the multiplications, and then the additions. Remember that any base raised to the exponent 0 is 1.

$$3 \times 10^2 + 0 \times 10^1 + 6 \times 10^0 = 3 \times 100 + 0 \times 10 + 6 \times 1 = 300 + 0 + 6 = 306$$

 YOUR TURN 4.1

Evaluate the following exponential expressions.

1. $3 \times 2^5 + 5 \times 8^2$
2. $5 \times 7^3 + 2 \times 7^2 + 5 \times 7^1 + 3 \times 7^0$
3. $1 \times 10^4 + 7 \times 10^3 + 4 \times 10^2 + 8 \times 10^1 + 8 \times 10^0$

Converting Hindu-Arabic Numerals to Expanded Form

When you see the number 738, and you speak the number out loud, what do you say? You probably said “seven hundred thirty-eight” while wondering what point could possibly be made by asking this. What you didn’t say was “seven, and three, and eight.” A pre-K student might say that. Which should make you wonder, why?

The reason is that you’ve been taught **place values**, or the positions of digits in a number that determine the values of those digits. You know that in a three-digit number, the first digit is hundreds, the second digit is tens, and the last digit is ones. These place values rely on powers of 10, which makes this system a **base 10 system**.

This sense of place value is what makes our system of numbers so useful. You’ve also been taught the **Hindu-Arabic numeration system**. This system, which uses the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9, and also employs place value based on powers of 10, is in use today.

Writing a number using these place values is writing them in **expanded form**. For a number with n digits, the expanded form is the first digit times 10 raised to one less than n , plus each following digit times 10 raised to one less than the previous power of 10. For example, the number 738 would be written as $7 \times 10^2 + 3 \times 10^1 + 8 \times 10^0$.

What about a four-digit number, like 5,825? Out loud, we’d say five thousand, seven hundred twenty-five. In expanded form, it would be $5 \times 10^3 + 8 \times 10^2 + 2 \times 10^1 + 5 \times 10^0$. Notice that the largest exponent is one less than the number of digits, and that the exponents go down by one as we move through the number.



PEOPLE IN MATHEMATICS

Aryabhata of Kusumapura and Brahmagupta

The Hindu-Arabic numeral system developed in India, and Aryabhata of Kusumapura is credited with the place value notation in the 5th century. However, the system wasn't as complete as it could be, until, roughly a century later, when Brahmagupta introduced the symbol for 0. The 0 is necessary to indicate that a given place value has been skipped, as in 4,098. In 4,098, the 10^2 power is skipped. Without such a symbol, 4,098 and 498 look similar. The value of both the place value notation and the introduction of the symbol 0 cannot be overstated, for math and the sciences.

EXAMPLE 4.2

Writing a Number in Expanded Form

Write the following in expanded form.

1. 563
2. 4,821
3. 903,786

✓ Solution

1. **Step 1:** Since there are three digits in 563, n is 3. So, this is the first digit times 10 raised to the power of 2, so we start with 5×10^2 .

Step 2: Then we add the next digit, 6, multiplied by 10 to a power one less than the previous, at which point we have $5 \times 10^2 + 6 \times 10^1$.

Step 3: Finally, the last digit is multiplied by 10 to the zeroth power and added to the previous. This results in $5 \times 10^2 + 6 \times 10^1 + 3 \times 10^0$.
2. **Step 1:** Since there are four digits in 4,821, n is 4. We multiply the first digit, 4, by 10 raised to the power of 3, which is 4×10^3 .

Step 2: Then we add the next digit, 8, multiplied by 10 to a power one less than the previous, at which point we have $4 \times 10^3 + 8 \times 10^2$.

Step 3: We continue to the next digit, lowering the exponent of 10 by one. Now we have $4 \times 10^3 + 8 \times 10^2 + 2 \times 10^1$.

Step 4: Finally, the last digit is multiplied by 10 to the zeroth power and added to the previous. This results in $4 \times 10^3 + 8 \times 10^2 + 2 \times 10^1 + 1 \times 10^0$.
3. Since there are six digits in 903,786, n is 6. So, we begin the process with 9 times 10 raised to the 5th power and continue through the numbers, reducing the exponent of 10 by one each time. This results in $9 \times 10^5 + 0 \times 10^4 + 3 \times 10^3 + 7 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$.



YOUR TURN 4.2

Write the following in expanded form.

1. 924
2. 1,279
3. 4,130,045

Converting Numbers in Expanded Form to Hindu-Arabic Numerals

Converting from expanded form back into a Hindu-Arabic numeral is the reverse process of expanding a number, and is equivalent to evaluating the exponential expression.

EXAMPLE 4.3**Converting a Number from Expanded Form to a Hindu-Arabic Numeral**

Convert the following into Hindu-Arabic numerals.

- $3 \times 10^2 + 4 \times 10^1 + 8 \times 10^0$
- $5 \times 10^3 + 0 \times 10^2 + 9 \times 10^1 + 9 \times 10^0$
- $6 \times 10^6 + 2 \times 10^5 + 0 \times 10^4 + 9 \times 10^3 + 1 \times 10^2 + 1 \times 10^1 + 7 \times 10^0$

 Solution

- Evaluating the expression results in: $3 \times 10^2 + 4 \times 10^1 + 8 \times 10^0 = 3 \times 100 + 4 \times 10 + 8 \times 1 = 300 + 40 + 8 = 348$
- Evaluating the expression results in:
 $5 \times 10^3 + 0 \times 10^2 + 9 \times 10^1 + 9 \times 10^0 = 5 \times 1000 + 0 \times 100 + 9 \times 10 + 9 \times 1 = 5000 + 0 + 90 + 9 = 5099$
- Evaluating the expression results in:
 $6 \times 10^6 + 2 \times 10^5 + 0 \times 10^4 + 9 \times 10^3 + 1 \times 10^2 + 1 \times 10^1 + 7 \times 10^0$
 $= 6 \times 1,000,000 + 2 \times 100,000 + 0 \times 10,000 + 9 \times 1,000 + 1 \times 100 + 1 \times 10 + 7 \times 1$
 $= 6,000,000 + 200,000 + 0 + 9,000 + 100 + 10 + 7$
 $= 6,209,117$

 YOUR TURN 4.3

Convert the following to Hindu-Arabic Numerals.

- $6 \times 10^2 + 2 \times 10^1 + 1 \times 10^0$
- $3 \times 10^3 + 2 \times 10^2 + 0 \times 10^1 + 3 \times 10^0$
- $4 \times 10^7 + 0 \times 10^6 + 6 \times 10^5 + 3 \times 10^4 + 0 \times 10^3 + 8 \times 10^2 + 9 \times 10^1 + 1 \times 10^0$

Check Your Understanding

- What is meant by a place value system?
- Evaluate the following exponential expression: $4 \times 8^2 + 2 \times 8^1 + 7 \times 8^0$.
- Express the following number in expanded form: 45,209.
- What number provides the value of a digit in our system of numeration?
- How are numerals and numbers different?
- Express as a Hindu-Arabic number: $6 \times 10^5 + 0 \times 10^4 + 1 \times 10^3 + 9 \times 10^2 + 4 \times 10^1 + 7 \times 10^0$.

**SECTION 4.1 EXERCISES**

- What does it mean for a system to be a place value system?
- In the system we use today, what number are the place values based on?
- How are numerals and numbers different?
- What relates numerals to numbers?

For the following exercises, evaluate the exponential expression.

- $3 \times 4^2 + 5 \times 2^3$
- $5 \times 6^3 + 7 \times 3^2$
- $7 \times 5^2 + 2 \times 4^5$
- $10 \times 11^2 + 7 \times 3^4$
- $5 \times 6^2 + 3 \times 6^1 + 4 \times 6^0$
- $4 \times 12^2 + 11 \times 12^1 + 2 \times 12^0$
- $14 \times 8^3 + 19 \times 5^4 + 2 \times 3^1$
- $8 \times 10^4 + 3 \times 5^3 + 9 \times 4^2$
- $1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

14. $5 \times 8^4 + 1 \times 8^3 + 0 \times 8^2 + 7 \times 8^1 + 3 \times 8^0$

For the following exercises, express the Hindu-Arabic number in expanded form.

15. 13

16. 25

17. 82

18. 99

19. 131

20. 408

21. 651

22. 3,901

23. 5,098

24. 12,430

For the following exercises, express the expanded number as a Hindu-Arabic number.

25. $3 \times 10^1 + 2 \times 10^0$

26. $5 \times 10^1 + 7 \times 10^0$

27. $2 \times 10^2 + 4 \times 10^1 + 9 \times 10^0$

28. $6 \times 10^2 + 0 \times 10^1 + 1 \times 10^0$

29. $1 \times 10^3 + 4 \times 10^2 + 4 \times 10^1 + 0 \times 10^0$

30. $7 \times 10^3 + 0 \times 10^2 + 1 \times 10^1 + 8 \times 10^0$

31. $6 \times 10^4 + 7 \times 10^3 + 0 \times 10^2 + 0 \times 10^1 + 0 \times 10^0$

32. $9 \times 10^4 + 8 \times 10^3 + 7 \times 10^2 + 3 \times 10^1 + 4 \times 10^0$

33. $7 \times 10^7 + 3 \times 10^6 + 4 \times 10^5 + 0 \times 10^4 + 4 \times 10^3 + 1 \times 10^2 + 5 \times 10^1 + 1 \times 10^0$

34. $8 \times 10^8 + 0 \times 10^7 + 4 \times 10^6 + 9 \times 10^5 + 9 \times 10^4 + 2 \times 10^3 + 2 \times 10^2 + 6 \times 10^1 + 0 \times 10^0$